HAT header and EEPROM

File: hat.kicad_sch

Power

File: power.kicad_sch

Phantom supply

File: phantom.kicad_sch

Codec

File: codec.kicad_sch

Panel switches

File: panel.kicad_sch

Input 1

PH_SW PH_SW1
PH_LED PH1_LED
LINE LINE1
ADC_P ADC1_P
ADC_M ADC1_M
PLUG PLUG1
PH_PRESENT PHP1

File: input.kicad_sch

Input 2

PH_SW PH_SW2
PH_LED PH2_LED
LINE LINE2
ADC_P ADC2_P
ADC_M ADC2_M
PLUG PLUG2
PH_PRESENT PHP2

File: input.kicad_sch

Output 1

DAC_P DAC1_P
DAC_M DAC1_M
PAD PAD1

File: output.kicad_sch

Output 2

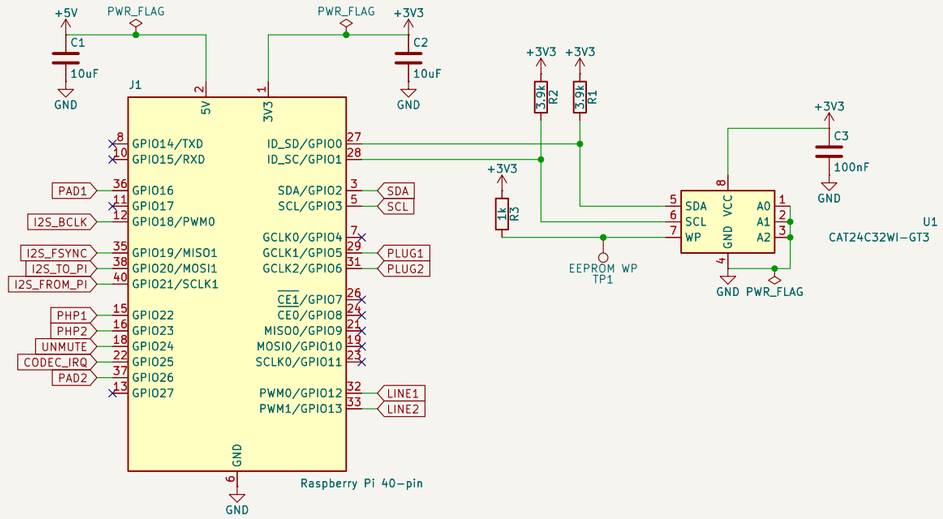
DAC_P DAC2_P
DAC_M DAC2_M
PAD PAD2

File: output.kicad_sch

Sheet: /
File: pedal-codec.kicad_sch

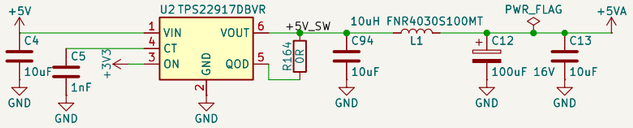
Title:

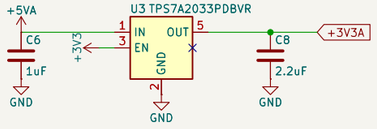
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/10

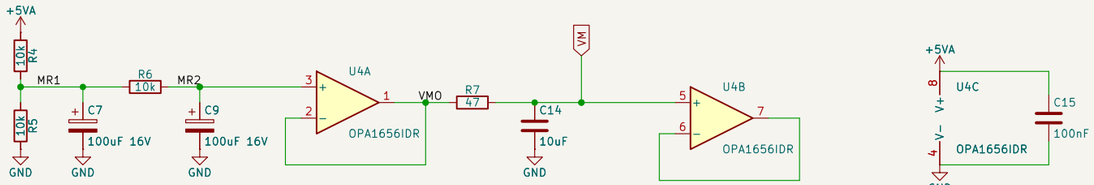


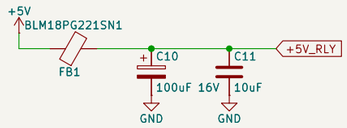
H1 M2.5 H2 M2.5 H3 M2.5 H4 M2.5

Sheet: /HAT header and EEPROM/		
File: hat.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
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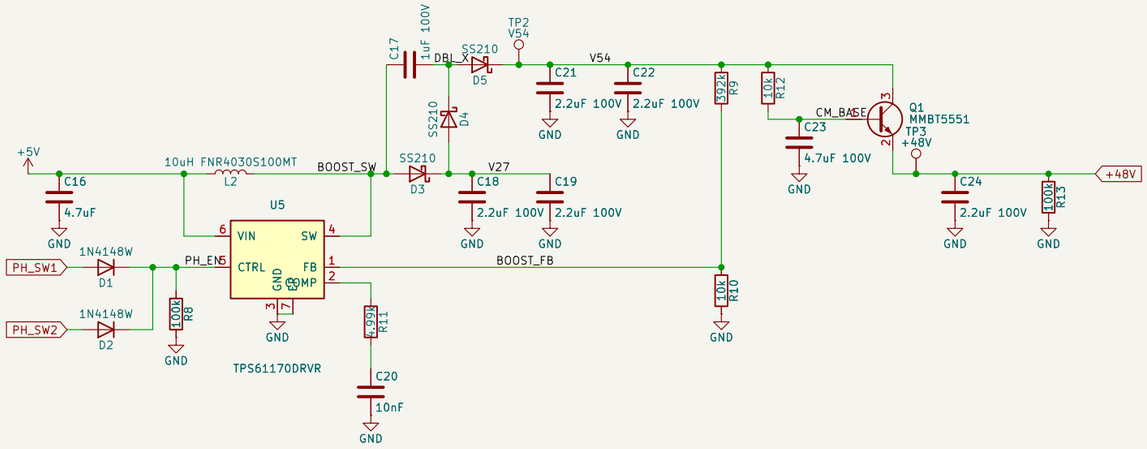




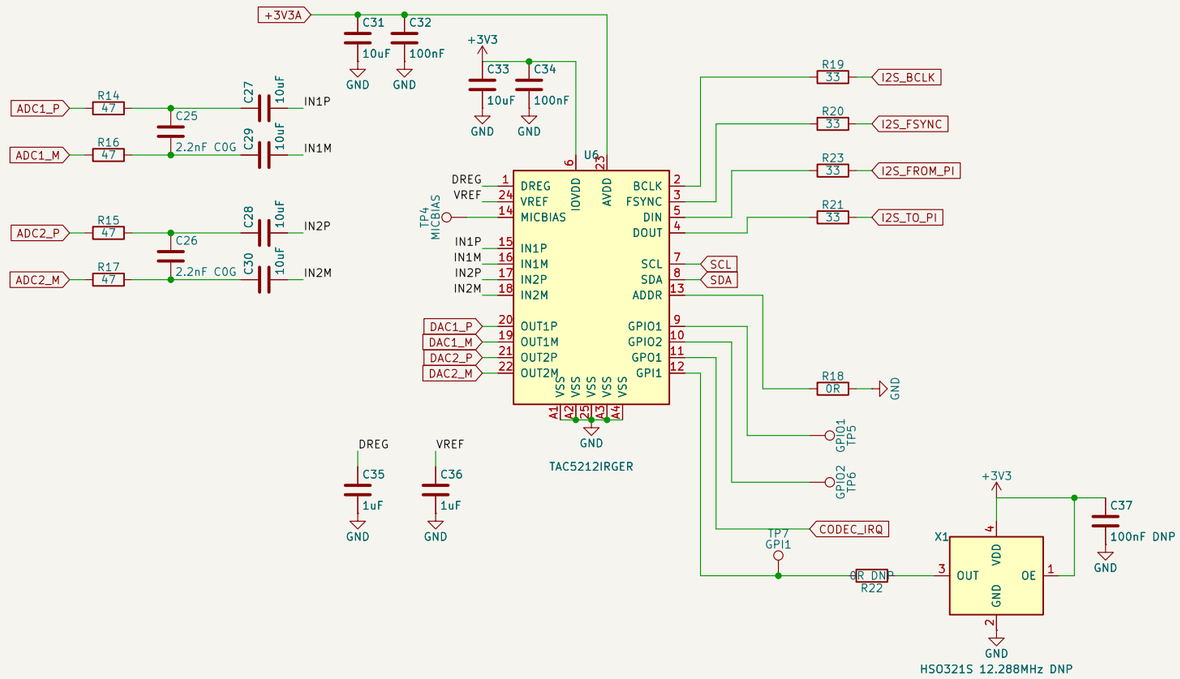
Sheet: /Power/
File: power.kicad_sch

Title:

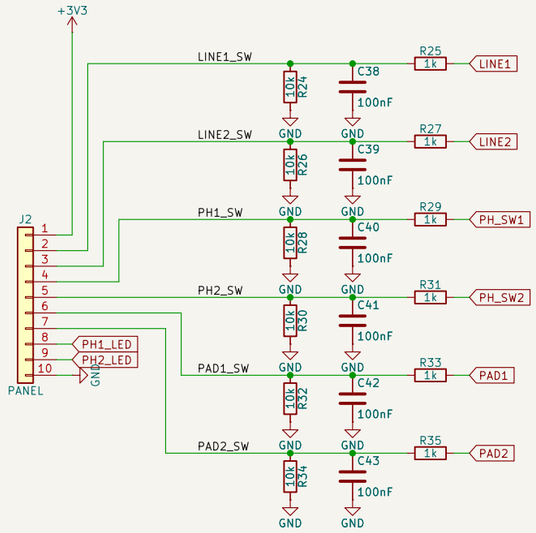
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 3/10

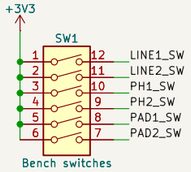


Sheet: /Phantom supply/ File: phantom.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 4/10

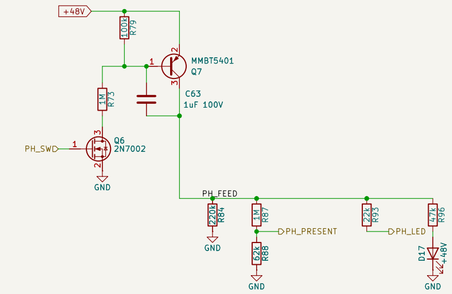


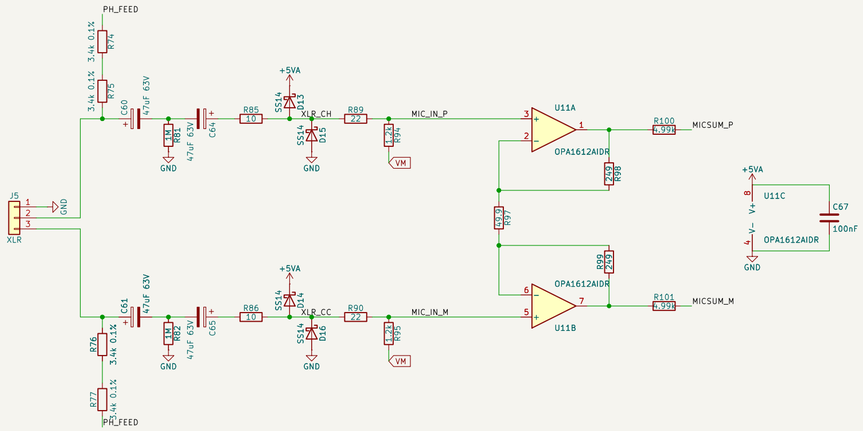
Sheet: /Codec/		
File: codec.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 5/10



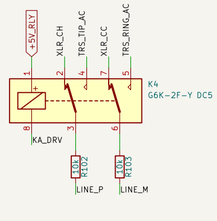


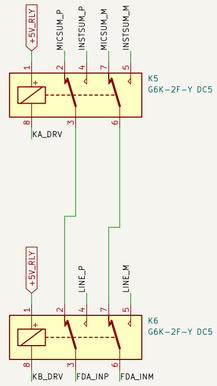
Sheet: /Panel switches/ File: panel.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 6/10

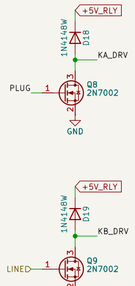

+48V
4.99k R79
MMBT5401
Q7
C63
1uF 100V
1M R75
PH_SW
Q6
2N7002
GND
PH_FEED
220k R84
1M R87
PH_PRESENT
PH_LED
2.21k R93
475 R96
620k R88
D17
24-48V

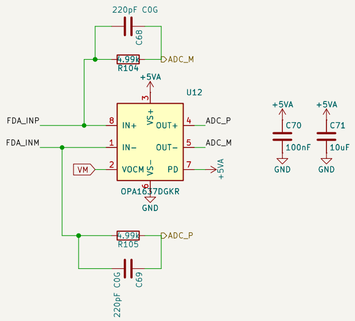

PH_FEED
3.44 0.1% R74
3.44 0.1% R75
C60
47uF 63V
J5
XLR
GND
1M R81
C64
47uF 63V
R85
10
SS14
D13
+5VA
XLR_CH
R89
22
MIC_IN_P
VM
U11A
OPA1612AIDR
R100
4.99k
MICSUM_P
220k R98
100k R97
R99
2.49k
U11C
C67
100nF
V+
V-
OPA1612AIDR
GND
R76
3.44 0.1% R77
PH_FEED
C61
47uF 63V
1M R82
C65
47uF 63V
R86
10
SS14
D14
XLR_CC
R90
22
MIC_IN_M
R91
4.99k
U11B
R101
4.99k
MICSUM_M

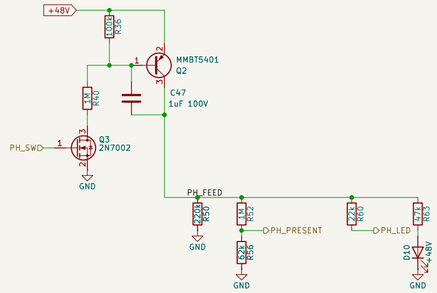
TRS_TIP_AC
J6
1/4in TRS
10uF
C58
GND
C59
100nF C0G
R83
1k
U10A
OPA2197IDR
R91
4.99k
INSTSUM_P
4.22k R82
1M R80
VM
GND
C62
10uF
TRS_RING_AC
100k R77
+3V3
10k R78
PLUG
U10B
OPA2197IDR
U10C
OPA2197IDR
+5VA
C66
100nF
R92
4.99k
INSTSUM_M

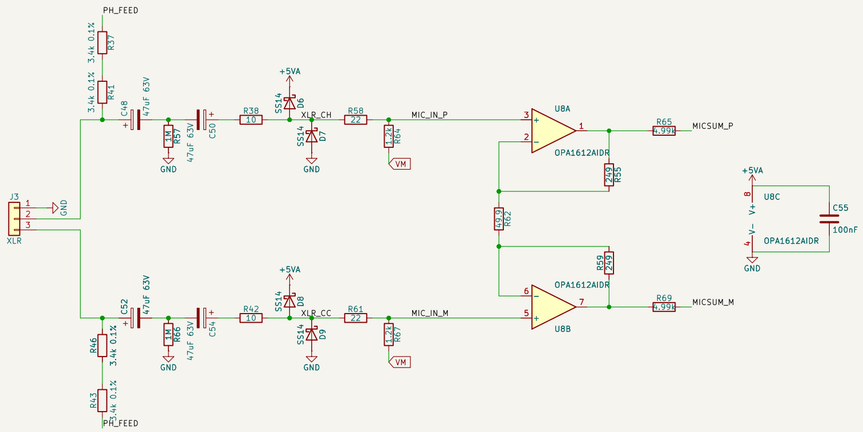

+5V_RLY
XLR_CH
TRS_TIP_AC
XLR_CC
TRS_RING_AC
K4
G6K-2F-Y DC5
KA_DRV
100 R102
100 R103
LINE_P
LINE_M

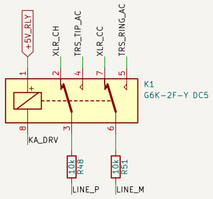

+5V_RLY
MICSUM_P
INSTSUM_P
MICSUM_M
INSTSUM_M
K5
G6K-2F-Y DC5
KA_DRV
+5V_RLY
LINE_P
LINE_M
K6
G6K-2F-Y DC5
KB_DRV
FDA_INP
FDA_INM

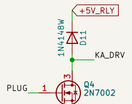

+5V_RLY
1N4148W
D18
KA_DRV
PLUG
Q8
2N7002
GND
+5V_RLY
1N4148W
D19
KB_DRV
LINED
Q9
2N7002
GND

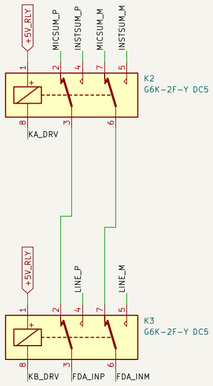

220pF C0G
C68
4.99k
R104
ADC_M
+5VA
U12
FDA_INP
FDA_INM
IN+
IN-
VOCM
VS+
VS-
OUT+
OUT-
PD
ADC_P
OPA1637DGKR
4.99k
R105
220pF C0G
C69
+5VA
C70
100nF
+5VA
C71
10uF
GND

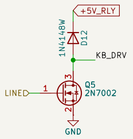


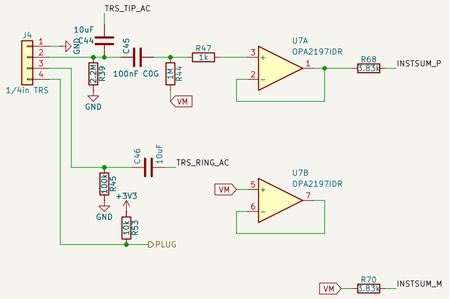


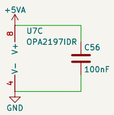


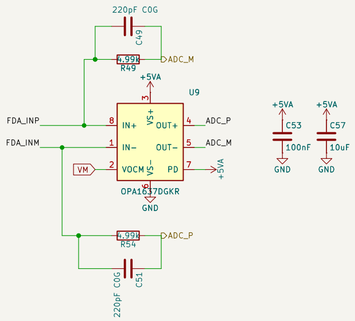












Sheet: /Input 2/
File: input.kicad_sch

Title:

Size: A2	Date:	**Rev:**
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DAC_PD
C72
10uF X7R
R112
1k
XP
R118
100k
VM
R107
91k 0.1%
R111
10k 0.1%
VM
U15A
TS5A23157DGSR
9
2
10
1
OPA1656IDR
U13A
3
2
1
R108
100
SS14
D20
SS14
D22
GND
+5VA
C74
47uF 63V
R120
1M
C76
47uF 63V
GND
R123
10
R126
100k
GND
GND
GND
R130
0R DNP
C82
10nF DNP
J8
1
2
3
OUT XLR
R134
10k
+5VA
GND
R125
10k
1.3k
R122
VM
PAD
1
2
3
Q11
2N7002
R129
10k
1.3k
R128
DAC_MD
C73
10uF X7R
R115
1k
XM
R119
100k
VM
R114
91k 0.1%
R117
10k 0.1%
VM
4
7
5
6
TS5A23157DGSR
U15B
6
5
7
U13B
OPA1656IDR
R109
100
SS14
D21
SS14
D23
+5VA
C75
47uF 63V
R124
10
R127
100k
GND
R121
1M
C77
47uF 63V
GND
GND
+5V_RLY
1N4148W
D24
VM
VM
K7
G6K-2F-Y DC5
1
2
4
7
5
8
3
6
XP
XM
UNMUTE
R132
100k
GND
1
Q10
2N7002
3
2
GND
XP
R106
9.09k
R110
7.5k
VM
OPA1656IDR
U14A
3
2
1
R116
7.5k
R113
9.09k
XM
C81
22uF 16V
R131
100
GND
J7
1
2
OUT unbalanced
R133
100k
GND
OPA1656IDR
U14B
VM
5
6
7
+5VA
U13C
8
V+
4
V-
C78
100nF
GND
OPA1656IDR
+5VA
U14C
8
V+
4
V-
C79
100nF
GND
OPA1656IDR
+5VA
U15C
8
VCC
GND
3
C80
100nF
GND
TS5A23157DGSR

Sheet: /Output 1/
File: output.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 9/10

DAC_PD
C83
10uF X7R
R136
1k
XP
R138
100k
VM
DAC_MD
C84
10uF X7R
R137
1k
XM
R139
100k
VM
U18A
TS5A23157DGSR
R144
91k 0.1%
R146
10k 0.1%
VM
R145
91k 0.1%
R147
10k 0.1%
VM
U18B
TS5A23157DGSR
+5VA
R148
10k
GND
Q13
2N7002
PAD
R151
10k
R152
10k
VM
U16A
OPA1656IDR
1.3k
R153
1.3k
R154
U16B
OPA1656IDR
R155
100
R156
100
+5VA
SS14
D26
SS14
D28
SS14
D27
SS14
D29
C86
47uF 63V
C87
47uF 63V
1M
R157
1M
R158
C89
47uF 63V
C90
47uF 63V
R159
10
R160
10
R161
100k
R162
100k
GND
GND
GND
GND
GND
GND
R163
0R DNP
C92
10nF DNP
J10
OUT XLR
+5V_RLY
1N4148W
D25
K8
G6K-2F-Y DC5
VM
VM
XP
XM
UNMUTE
R135
100k
Q12
2N7002
GND
GND
XP
XM
R140
9.09k
R141
9.09k
VM
R142
7.5k
R143
7.5k
OPA1656IDR
U17A
C85
22uF 16V
R149
100
R150
100k
GND
J9
OUT unbalanced
VM
OPA1656IDR
U17B
+5VA
U16C
C88
100nF
GND
OPA1656IDR
+5VA
U17C
C91
100nF
GND
OPA1656IDR
+5VA
U18C
VCC
GND
C93
100nF
GND
TS5A23157DGSR

Sheet: /Output 2/
File: output.kicad_sch

Title:

Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 10/10